

Causal ML: prediction accuracy is not enough

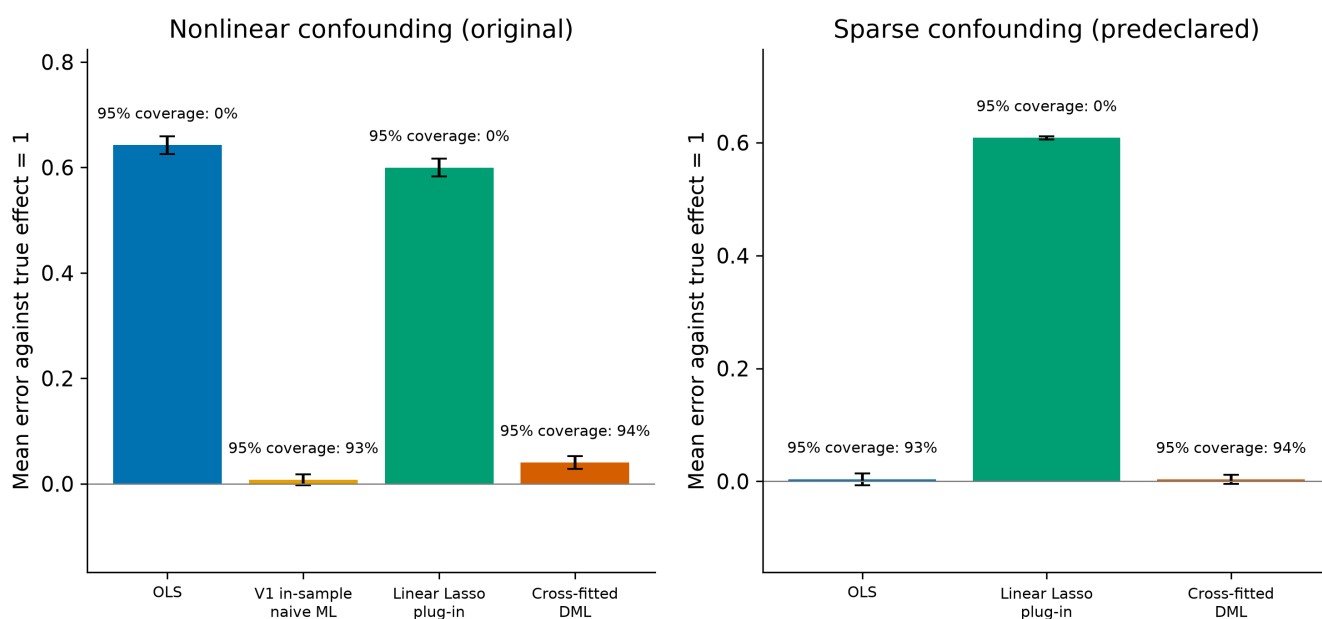
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Question

When can prediction-focused machine learning distort an estimated treatment effect, and does cross-fitting help?

Method

We compare effect estimators in two seeded experiments where the true effect is known. We retain the original nonlinear design and add a predeclared sparse-confounding design with two hundred candidate controls and Lasso nuisance estimates to illustrate regularization bias under strong confounding. We measure bias, root mean squared error and confidence-interval coverage, alongside the unchanged public-data replication.



100 repetitions each; bars: 95% Monte Carlo uncertainty for mean bias. Coverage is for effect confidence intervals. DGP1: boosting DML, linear Lasso plug-in; DGP2: Lasso nuisances. Original in-sample comparator retained.

Sources: original seeded simulations; public 1991 SIPP extract via DoubleML, private retention. Author calculations; results/dml_two_dgps.csv and unchanged pension.csv.

Three findings for the decision-maker

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1. Original nonlinear design: V1 in-sample naive ML bias is 0.008, versus 0.600 for the linear Lasso plug-in; OLS bias is 0.642, while cross-fitted DML bias is 0.040 with 94% coverage.
2. With 200 candidate confounders, naive plug-in bias is 0.609 and coverage 0%; cross-fitted DML bias is 0.003, RMSE 0.042 and coverage 94%.
3. The unchanged retirement-account replication estimates \$8,172 more net financial assets for eligible people (95% interval \$5,795 to \$10,549), assuming measured controls remove confounding.

Limitations

- These illustrative designs do not establish universal estimator rankings.
- The nuisance learners and dimensions differ between designs.
- Naive plug-in intervals ignore nuisance-estimation uncertainty.
- A hundred repetitions give imprecise coverage estimates.
- The real-data estimate depends on untestable confounding assumptions.

Detailed assumptions, diagnostics and source lineage: [README](#) [Technical notes](#) and [report/report.md](#).